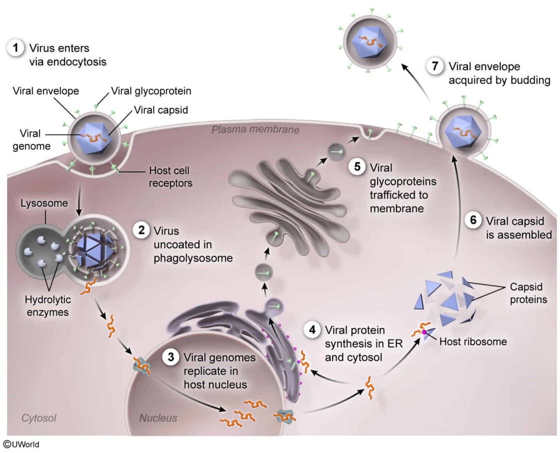
An epithelial cell in the upper respiratory tract was infected with an enveloped virus, such as influenza virus. Upon entering the host cell, the viral life cycle was initiated and the host cell machinery was used to synthesize new viral components. Given this information, which of the following statements is LEAST likely to be true?

- A. Inhibition of protein synthesis in the cytosol would limit the assembly of new viruses. [14%]
- B. Disruption of trafficking between the Golgi apparatus and cell surface would result in the production of fewer viral glycoproteins. [11%]
- ✔ C. Inhibition of lysosomal trafficking would prevent new viruses from being released. [61%]
- D. Disruption of protein synthesis in the rough endoplasmic reticulum would inhibit the release of new viruses. [12%]

Correct
60% answered correctly

Explanation:

Viral assembly requires host protein synthesis and trafficking machinery



Viruses are intracellular parasites that require host cell machinery to replicate. Once a virus enters a host cell, the viral genome replicates and viral proteins must be synthesized to assemble new viruses. The virus uses the host cell's protein synthesis and trafficking machinery to synthesize and correctly position viral proteins for assembly.

Viral assembly begins once all viral components are synthesized and properly located. Enveloped viruses acquire a plasma membrane–derived envelope containing viral proteins while exiting the host cell. These viral proteins are synthesized and trafficked to the plasma membrane using the host's endomembrane system (ie, endoplasmic reticulum [ER], Golgi apparatus, transport vesicles).

The question asks which component of host cell machinery is *least* involved with the assembly of an enveloped virus. The virus must use host machinery to synthesize viral proteins in the cytosol. Because the question specifies an enveloped virus, viral surface

proteins must also be trafficked to the plasma membrane through the host endomembrane system.

Therefore, **inhibiting** lysosomal **trafficking** would have the **least impact** on the assembly of a new virus, because lysosomes function primarily in the degradation of macromolecules in the cell, *not* protein synthesis or trafficking to the cell membrane. Lysosomes do have a role in viral *entry* by endocytosis, but the virus in question has already entered the host cell.

The other choices *would* affect the assembly of a new enveloped virus in an infected cell.

(Choice A) Viral mRNA is translated by host ribosomes in the cytosol to create viral proteins needed to assemble viruses. Therefore, inhibiting protein synthesis in the cytosol *would* limit viral assembly.

(Choices B and D) Enveloped viruses exit the host cell by budding from the host plasma membrane, creating a viral envelope studded with viral proteins. If protein trafficking between the Golgi apparatus and cell surface or protein synthesis at the rough ER was inhibited, viral proteins would *not* arrive at the cell surface to become part of the viral envelope, limiting viral assembly.

Educational objective:
Viruses are intracellular parasites that lack the metabolic machinery to replicate on their own. Viruses must utilize host cell machinery for genome replication and protein synthesis. Viral proteins are synthesized via host ribosomes in the cytosol and/or trafficked through the endomembrane system to the proper location prior to assembly.

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